Sai Mahendra Penumakula

Martur, AP · +91-900-00-15697· saimahipenumakula@gmail.com · LinkedIn · GitHub · Portfolio

SUMMARY

Backend Software Engineer with 3+ years of experience designing and delivering secure, production-ready **Java/Spring microservices and RESTful APIs**. Hands-on with **Spring Framework (Spring MVC, Spring Boot, JAX-RS, Spring Data JPA for persistence, Hibernate**, and building microservices-based systems using **REST**. Implemented authentication and authorization with **Spring Security (OAuth2, JWT)**. Experienced in **AWS cloud environments (EC2, S3, RDS, Lambda)** and CI/CD automation. **AWS Certified Solutions Architect – Associate**, with hands-on experience modernizing legacy systems, automating document workflows, and integrating external systems using scalable, cloud-ready architectures.

SKILLS

- **Languages & Platforms**: Java (8 / 11 / 17), J2EE, JavaScript, TypeScript
- **Spring & Server-side**: Spring Boot, Spring MVC, Spring Data JPA, Spring Security (JWT, OAuth2), Hibernate, Microservices
- **APIs & Integration**: RESTful APIs, JSON, XML, Swagger / Open API, External API integrations, JSON / XML
- **Persistence & Data**: Relational: SQL Server (Zyter), MySQL (exposure), PostgreSQL, NoSQL, ORM: Hibernate / JPA
- **Frontend**: Zyter: Angular (TypeScript — Angular 10 & 17 exposures), React (JavaScript), JSP, Thyme leaf, HTML, CSS
- **Containers, Infra & CI/CD:** AWS, Jenkins, GitHub Actions, pipeline automation (containerized builds, Maven)
- **Testing, Build & API Tools**: JUnit, Mockito, Jasmine, Karma, Swagger, Postman, OpenAPI (3.0)
- **Security & Best Practices**: OAuth2, JWT, authentication / authorization best practices, secure coding
- **Tools & IDEs**: IntelliJ IDEA, VS Cod, Eclipse, Spring Tool Suite (STS), Git, Bitbucket, GitHub, Jira, Confluence

EXPERIENCE

Software Engineer – Zyter|Trucare **Denton, TX · Mar 2024 – Sep 2025**

- Developed and maintained backend services for a large monolithic healthcare platform using **Java 8, 11 17, Spring Boot, and JAX-RS (REST Easy)**.
- Owned backend functionality for the **UM Authorization module,** supporting automated generation of patient and authorization letters.
- Implemented document automation using **Aspose.Words**, enabling dynamic **Word-to-PDF generation using XML-based** templates.
- Led a major **Spring Boot upgrade (2.7 → 3.5.3)**, including **Jakarta (javax → jakarta) migration** and dependency modernization.
- Modernized application security using **Spring Security 6,** implementing **OAuth2 and JWT-based** authentication.
- Integrated **springdoc-openapi to provide OpenAPI 3.0 documentation** and Swagger UI.
- Enhanced outbound service communication by updating **Feign client authentication** handling.
- Wrote **unit and integration tests using JUnit and Mockito**; resolved production and post-upgrade regression issues.
- Collaborated with QA, product, and support teams in an **Agile/Scrum environment**.

Software Engineer – Singtel Telecommunications Limited **Chennai, India · May 2020 – Oct 2021**

- Developed and enhanced **Java/Spring Boot microservices** supporting high-volume telecom billing systems.
- Built **REST APIs for Subscription Management and Invoice & Payments services**.
- Integrated **microservices with centralized Billing & Revenue Management (BRM) systems via Oracle Service Bus (OSB)**.
- Implemented batching, background processing, and caching strategies to support large-scale billing workloads.
- Worked on **invoice generation workflows**, assembling **BRM-calculated charges** into final billing outputs.
- Supported CI/CD pipelines using Jenkins and performed unit testing with JUnit.
- Collaborated closely with cross-functional teams in an Agile environment.

PROJECTS

PDF Automation & Reporting (TruCare) — Zyter|Trucare

- Implemented **Aspose-based PDF generation** and templating pipeline integrated into Spring services; automated patient correspondence and authorization letters using **XML merge-tags in Word templates**.

Spring Boot Upgrade & Security Modernization — Zyter|Trucare

- Upgraded monolith from **Spring Boot 2.7 → 3.5.3**; refactored security to use **SecurityFilterChain**, migrated javax. → jakarta., integrated springdoc-openapi, and standardized **JWT/OAuth2 flows**.

CareWebQI Integration – GET to POST Migration (Zyter)

- Migrated external system integration from **GET-based URL parameters to secure POST requests** to handle large payloads.
- Preserved parameter order and added backend-generated security hashing for request integrity.
- Improved security by eliminating sensitive data exposure in URLs and enabling reliable transmission of 50+ parameters.

Telecom Billing Optimization System — Singtel

- Architected Java microservices using Hibernate/JPA with Oracle DB and MongoDB for event storage; integrated **Kafka** for decoupled billing events and used containerization for environment parity; improved throughput and latency through partitioning and query optimization.

Portfolio Projects — GitHub

- Developed multiple full-stack and backend-focused projects — including **Angular-Staff, Spring-React-Shop, Quiz Forge Microservices, and others** — integrating a broad set of enterprise technologies across Java, J2EE, Spring Boot, microservices, REST, JSP, Thyme leaf, Hibernate/JPA, Swagger/OpenAPI, PostgreSQL/MySQL, and modern frontend stacks (React, Angular). These projects collectively demonstrate end-to-end system design, API development, secure service integration, and scalable backend architecture.

CERTIFICATIONS

• AWS Certified Solutions Architect – Associate (SAA-C03). Badge: https://www.credly.com/badges/bae4a082-dfc7-479a-a2a1-d80fa82dc9c2/public_url

EDUCATION

MS- Computer Science – Auburn University Montgomery **Jan 2022 – May 2023**

BE- Electronics & Communication Engineering – Sathyabama University - Chennai **June 2014 – Nov 2018**